

Table 7a Example 1 Solution 1

House	#1	#2	#3	#4	#5
Color	Red	Green	Yellow	White	Blue
Name	Tom	An	Nancy	Kay	Mike
Job	Doctor	Teacher	Artist	Writer	Nurse
Sport	Running	Basketball	Volleyball	Football	Swimming
Book	Fantasy	Science	Mystery	History	Romance
Instrument	Violin	Piano	Drums	Flute	Guitar
Total	32	36	39	34	34

Table 7b Example 1 Solution 2

House	#1	#2	#3	#4	#5
Color	Red	Green	Yellow	White	Blue
Name	Tom 3	Mike 4	Nancy 5	Kay	An
Job	Doctor	Teacher	Artist	Writer	Nurse
Sport	Running	Basketball	Volleyball	Football	Swimming
Book	Fantasy	Science	Mystery	History	Romance
Instrument	Violin	Piano	Drums	Flute	Guitar
Total	32	38	39	34	32

Table 7c Example 1 Solution 3

House	#1	#2	#3	#4	#5
Color	Red	Green	Yellow	White	Blue
Name	Tom	Mike	Nancy	An	Kay
Job	Doctor	Teacher	Artist	Writer	Nurse
Sport	Running	Basketball	Volleyball	Football	Swimming
Book	Fantasy	Science	Mystery	History	Romance
Instrument	Violin	Piano	Drums	Flute	Guitar
Total	32	38	39	33	33

No coding, no tools—just direct deduction to solve the following problem.

There are five houses of different colors with different jobs; they love different books, sports, and instruments. The entities and constraints are as follows:

Color: {Blue, Green, Red, White, Yellow}

Name: {An, Kay, Mike, Nancy, Tom}

Job: {Artist, Doctor, Nurse, Teacher, Writer}

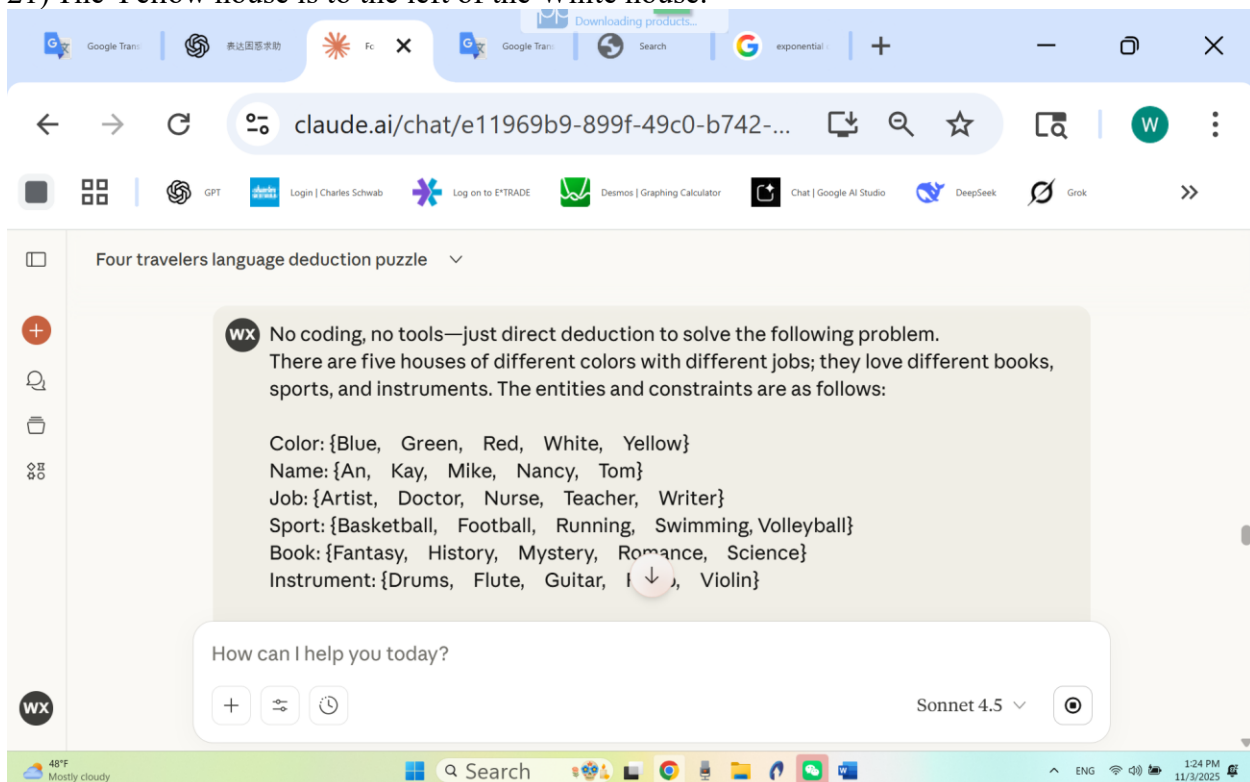
Sport: {Basketball, Football, Running, Swimming, Volleyball}

Book: {Fantasy, History, Mystery, Romance, Science}

Instrument: {Drums, Flute, Guitar, Piano, Violin}

- 1) The Doctor is next to the Teacher.
- 2) Tom is the Running lover.
- 3) The Romance lover is in house 1 or house 5.

- 4) There exist two instrument groups—string and non-string (S and $\sim S$)—such that ball sport enthusiasts (B) and non-ball sport enthusiasts ($\sim B$) are strictly assigned to different groups.
- 5) Football lover is somewhere to the right of the Basketball lover.
- 6) The Writer is anywhere between the Nurse and the Doctor.
- 7) The Romance lover is a Swimming lover.
- 8) The Basketball lover lives in the Green house.
- 9) The Drums owner is anywhere to the right of the Piano owner.
- 10) For each house, add the number of letters in its color, job, book, sport, and instrument. The totals form the set $S = \{s_1, s_2, s_3, s_4, s_5\}$. There exist at least two houses with the same total.
- 11) The History lover is anywhere to the right of the Drums owner.
- 12) The Fantasy lover is next to the Science lover.
- 13) Even-letter names not adjacent
- 14) The Fantasy lover is at one of the ends.
- 15) The Red house is somewhere to the left of the Basketball lover.
- 16) The Writer is anywhere to the right of the Drums owner.
- 17) Nancy is anywhere between the Green house and Kay, in this sequence.
- 18) The Blue house is not next to the Volleyball lover.
- 19) The Piano owner is anywhere between the Doctor and the Guitar owner, in this sequence.
- 20) Tom is in house 1 or house 5.
- 21) The Yellow house is to the left of the White house.



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- 1) The Doctor is next to the Teacher.
- 2) Tom is the Running lover.
- 3) The Romance lover is in house 1 or house 5.
- 4) There exist two instrument groups—string and non-string (S and ~S)—such that ball sport enthusiasts (B) and non-ball sport enthusiasts (~B) are strictly assigned to different groups.
- 5) Football lover is somewhere to the right of the Basketball lover.
- 6) The Writer is anywhere between the Nurse and the Doctor.
- 7) The Romance lover is a Swimming lover.
- 8) The Basketball lover lives in the Green house.
- 9) The Drums owner is anywhere to the right of the Piano owner.

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- 8) The Basketball lover lives in the Green house.
- 9) The Drums owner is anywhere to the right of the Piano owner.
- 10) For each house, add the number of letters in its color, job, book, sport, and instrument. The totals form the set $S = \{s_1, s_2, s_3, s_4, s_5\}$. There exist at least two houses with the same total.
- 11) The History lover is anywhere to the right of the Drums owner.
- 12) The Fantasy lover is next to the Science lover.
- 13) Even-letter names not adjacent
- 14) The Fantasy lover is at one of the ends.
- 15) The Red house is somewhere to the left of the Basketball lover.
- 16) The Writer is anywhere to the right of the Drums owner.
- 17) Nancy is anywhere between the Green house and Kay, in this sequence.

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17) Nancy is anywhere between the Green house and Kay, in this sequence.
18) The Blue house is not next to the Volleyball lover.
19) The Piano owner is anywhere between the Doctor and the Guitar owner, in this sequence.
20) Tom is in house 1 or house 5.
21) The Yellow house is to the left of the White house.

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I'll solve this step-by-step using logical deduction.

Step 1: Establish initial constraints

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Step 1: Establish initial constraints

From (2) and (20): Tom loves Running and is in house 1 or 5.
From (3) and (7): Romance lover is in house 1 or 5 AND loves Swimming.
From (14): Fantasy lover is in house 1 or 5.

Since Tom loves Running (not Swimming), Tom $\neq$ Romance lover.
So if Tom is in house 1, Romance must be in house 5 (and vice versa).
This means Fantasy is in the same house as either Tom or Romance lover.

From (12): Fantasy lover is next to Science lov

- If Fantasy is in house 1, Science is in house 2.

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This means Fantasy is in the same house as either Tom or Romance lover.

From (12): Fantasy lover is next to Science lover.

- If Fantasy is in house 1, Science is in house 2.
- If Fantasy is in house 5, Science is in house 4.

Step 2: Determine Tom's position

If Tom (Running) is in house 1:

- Fantasy is in house 1, Science in house 2
- Romance (Swimming) is in house 5

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- Romance (Swimming) is in house 5

If Tom (Running) is in house 5:

- Fantasy is in house 5, Science in house 4
- Romance (Swimming) is in house 1

Let me test the first scenario.

Step 3: Test Tom in house 1

- House 1: Tom, Running, Fantasy

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- House 1: Tom, Running, Fantasy
- House 2: Science
- House 5: Swimming, Romance

From (8): Basketball lover lives in Green house.
 From (15): Red house is left of Basketball (Green) lover.
 From (5): Football is right of Basketball.

This means: Red ... Green (Basketball) ... Football

From (21): Yellow is left of White.

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FROM (21): YELLOW IS LEFT OF WHITE.

Step 4: Determine sports distribution

We have: Running (house 1), Swimming (house 5), Basketball, Football, Volleyball.
 From (5): Basketball is left of Football.
 From (8): Basketball is in Green house.

Possible positions for Basketball: houses 2, 3, or 4 (not 1 or 5).
 If Basketball is in house 2, Football can be in 3, 4, or 5.
 But house 5 has Swimming, so Football is in 3 or 4.

Step 5: Apply constraint (4) - Instruments and sports

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Step 5: Apply constraint (4) - Instruments and sports

String instruments: Guitar, Piano, Violin
 Non-string: Drums, Flute

Ball sports (B): Basketball, Football, Volleyball
 Non-ball sports (~B): Running, Swimming

Constraint (4): Ball sport lovers and non-ball sport lovers must be in different instrument groups.

- All B people have string instruments OR all ~B people have non-string instruments

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groups.

- All B people have string instruments OR all B people have non-string instruments
- All ~B people have the opposite type

Since there are 3 ball sports and 2 non-ball sports, and 3 string + 2 non-string:

- Ball sports → String instruments (Guitar, Piano, Violin)
- Non-ball sports → Non-string instruments (Drums, Flute)

So: Tom (Running) and house 5 person (Swimming) have Drums and Flute.
 Basketball. Football. Volleyball lovers have Guitar. Piano. Violin.

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30. Tom (Running) and house 5 person (Swimming) have Drums and Flute.
Basketball, Football, Volleyball lovers have Guitar, Piano, Violin.

Step 6: Apply instrument ordering constraints

From (9): Drums owner is right of Piano owner.
From (19): Piano owner is between Doctor and Guitar owner (Doctor ... Piano ... Guitar).
Since Piano is a string instrument, Piano owner loves a ball sport.
Since Drums is right of Piano, and Drums is non-string (non-ball sport):

- Piano owner (ball sport) is in houses 1-4
- Drums owner (Running or Swimming) is to the right

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- Drums owner (Running or Swimming) is to the right

Tom (Running, house 1) or house 5 (Swimming) has Drums.
If Tom has Drums, then Piano is impossible (nothing to the left).
So house 5 has Drums, and Tom (house 1) has Flute.

From (11): History lover is right of Drums owner.
Since Drums is in house 5, this is impossible!

Step 7: Test Tom in house 5

- House 5: Tom, Running, Fantasy

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- House 5: Tom, Running, Fantasy
- House 4: Science
- House 1: Swimming, Romance

Now Drums and Flute go to houses 1 and 5 (Running and Swimming).

From (9): Drums is right of Piano.

If house 1 has Drums, Piano must be... impossible (nothing left).

So house 5 (Tom) has Drums, house 1 has Flute.

From (11): History is right of Drums owner (house 5).

This is impossible!

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From (11): History is right of Drums owner (house 5).

This is impossible!

Step 8: Reconsider - Drums must not be at an end

Let me reconsider the ball/non-ball sport assignment.

What if: Non-ball sports → String, Ball sports → Non-string?

- Running (Tom), Swimming: Guitar, Piano, Violin
- Basketball, Football, Volleyball: Drums, Flute

But there are only 2 non-string instruments for 3 ball sports. Impossible.

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but there are only 2 non-string instruments for 3 ball sports. impossible.

So my original assignment was correct. But then constraints (9) and (11) create issues with Drums at house 5.

Step 9: Re-examine the constraints more carefully

Let me reconsider whether my sport-instrument mapping is necessarily correct.

Actually, constraint (4) says ball sports and non-ball sports are in DIFFERENT groups, but doesn't specify which group gets which instrument type. However, with 3 ball sports and 2 non-ball sports, and 3 string + 2 non-string instruments, the only way to "strictly assign" them to different groups is:

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them to different groups is:

- 3 ball sports → 3 string instruments
- 2 non-ball sports → 2 non-string instruments

Now, with Tom in house 1 (Running, Fantasy) and house 5 (Swimming, Romance):
Both Running and Swimming need Drums or Flute.

From (9): Drums right of Piano.
From (11): History right of Drums.

If Drums is in house 1, constraints (9) and (11) fail.

If Drums is in house 5, constraint (11) fails.

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If Drums is in house 1, constraints (9) and (11) fail.
 If Drums is in house 5, constraint (11) fails.
 So Drums must be in houses 2, 3, or 4.
 But Drums owner must love Running or Swimming (non-ball sports).
 Running is in house 1, Swimming is in house 5.
 Contradiction!

Step 10: Re-examine Tom's position assumption

Let me try Tom in house 5 again more careful

- House 5: Tom, Running, Fantasy

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- House 5: Tom, Running, Fantasy
- House 4: Science
- House 1: Swimming, Romance

For Drums (non-ball sport instrument) to be in houses 2-4:
 One of Running or Swimming must be in houses 2-4. But Running is in house 5, Swimming in house 1. Contradiction again.

Step 11: Question the Fantasy position

Wait - I assumed Fantasy must be with Tom or Romance. Let me reconsider.

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Wait - I assumed Fantasy must be with Tom or Romance. Let me reconsider.

From (14): Fantasy is at an end (house 1 or 5).
From (3): Romance is in house 1 or 5.
From (7): Romance lover loves Swimming.
From (2) and (20): Tom loves Running and is in house 1 or 5.

These are independent! Fantasy doesn't have to be with Tom or Romance.

Case A: Tom in house 1, Fantasy in house 1

- House 1: Tom, Running, Fantasy

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- House 1: Tom, Running, Fantasy
- House 2: Science (from constraint 12)
- Romance + Swimming: house 5

Case B: Tom in house 1, Fantasy in house 5

- House 1: Tom, Running
- House 4: Science (Fantasy in 5, next to Science in 4)
- House 5: Fantasy
- Romance + Swimming: house 5

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- House 1: Tom, Running, Fantasy
- House 2: Science (from constraint 12)
- Romance + Swimming: house 5

Case B: Tom in house 1, Fantasy in house 5

- House 1: Tom, Running
- House 4: Science (Fantasy in 5, next to Science in 4)
- House 5: Fantasy
- Romance + Swimming: house 5

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- Romance + Swimming: house 1
- So house 1: Fantasy, Romance, Swimming

Case D: Tom in house 5, Fantasy in house 5

- House 5: Tom, Running, Fantasy
- House 4: Science
- Romance + Swimming: house 1

Let me focus on **Case C** as it seems most promising for instrument constraints.

Step 12: Develop Case C

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Step 12: Develop Case C

- House 1: Fantasy, Romance, Swimming, Flute
- House 2: Science
- House 5: Tom, Running, Drums

From (11): History is right of Drums (house 5). Impossible!

Try Case D:

- House 1: Romance, Swimming, Flute

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- House 1: Romance, Swimming, Flute
- House 4: Science
- House 5: Tom, Running, Fantasy, Drums

From (11): History is right of house 5. Impossible!

Try Case B:

- House 1: Tom, Running, Flute
- House 4: Science
- House 5: Fantasy, Romance, Swimming, Drums

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- House 5: Fantasy, Romance, Swimming, Drums

From (11): History right of house 5. Impossible!

Try Case A:

- House 1: Tom, Running, Fantasy, Flute (or Drums)
- House 2: Science
- House 5: Romance, Swimming, Drums (or Flute)

From (9): Drums right of Piano.
From (11): History right of Drums.

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From (11): History right of Drums.

If house 1 has Drums: Piano impossible, History impossible.
If house 5 has Drums: History impossible.

Step 13: Drums must be in middle houses

This suggests Running or Swimming is NOT at the ends!

But Tom (Running) is in house 1 or 5 (constraint 20).
And Romance (Swimming) is in house 1 or 5 (constraint 3+7).

Unless... let me check if there's another person with Swimming?

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Unless... let me check if there's another person with Swimming?

No, constraint (7) specifically says Romance lover is Swimming lover. And there's only one Swimming.

Step 14: Re-examine constraint (4)

Let me reread constraint (4): "There exist two instrument groups—string and non-string (S and $\sim$ S)—such that ball sport enthusiasts (B) and non-ball sport enthusiasts ($\sim$ B) are strictly assigned to different groups."

Maybe I misinterpreted. Perhaps it means:

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Maybe I misinterpreted. Perhaps it means:

- IF someone likes ball sports, they're in group S (or $\sim$ S)
- IF someone likes non-ball sports, they're in group $\sim$ S (or S)
- But the ASSIGNMENT of which group gets string vs non-string is what we determine?

Actually, re-reading more carefully: "ball sport enthusiasts and non-ball sport enthusiasts are strictly assigned to different groups"

This means:

- All ball sport lovers are in ONE group (either all string or all non-string)

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- All ball sport lovers are in ONE group (either all string or all non-string)
- All non-ball sport lovers are in the OTHER group

Given 3 ball + 2 non-ball = 5 people, and 3 string + 2 non-string = 5 instruments:

- Option 1: Ball (3) → String (3), Non-ball (2) → Non-string (2)
- Option 2: Ball (3) → Non-string... wait, only 2 non-string. Impossible.

So Option 1 is correct.

Step 15: Consider if Running/Swimming ca ↓ in middle

Wait, what if I'm wrong about Tom and Romance positions?

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Wait, what if I'm wrong about Tom and Romance positions?

Let me reread (20): "Tom is in house 1 or house 5."

This is clear. Tom must be in 1 or 5.

But maybe Running doesn't have to be with Tom? No, (2) says "Tom is the Running lover."

And (3) says Romance lover is in house 1 or 5.

And (7) says Romance lover is Swimming lover.

So Swimming is definitely in house 1 or 5.

Step 16: Could there be an error in my instrument deduction?

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Step 16: Could there be an error in my instrument deduction?

Let me trace through again:

- Running and Swimming are non-ball sports
- By constraint (4), they must have non-string instruments (Drums, Flute)
- Basketball, Football, Volleyball are ball sports
- They must have string instruments (Guitar, Piano, Violin)

↓

This seems airtight.

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This seems airtight.

But then Drums and Flute are in houses {1, 5} (where Running and Swimming are).

From (9): Drums is right of Piano.

From (11): History is right of Drums.

If Drums is in house 1: Piano must be in house 0 or less. Impossible.

If Drums is in house 5: History must be in house 6 or more. Impossible.

This is a true contradiction!

Step 17: Reconsider constraint interpretation ↓

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Step 1: Reconsider constraint interpretations

Let me check if "anywhere to the right" means "immediately to the right" or "somewhere to the right."

Looking at constraint (5): "Football lover is somewhere to the right of the Basketball lover."

Looking at constraint (6): "The Writer is anywhere between the Nurse and the Doctor."

These seem to indicate "somewhere" not "immediately."

So "Drums owner is anywhere to the right of Piano owner" means Drums's house number > Piano's house number.

Similarly, "History lover is anywhere to the right of Drums owner" means History's house >

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Similarly, "History lover is anywhere to the right of Drums owner" means History's house > Drums's house.

If Drums is in house 5, then History must be in house > 5. Impossible.

If Drums is in house 1, then Piano must be in house < 1. Impossible.

So Drums must be in houses 2, 3, or 4.

But Drums belongs to a non-ball sport lover (Running or Swimming). Running is in house 1 or 5. Swimming is in house 1 or 5.

This is impossible!

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This is impossible!

Step 18: Final reconsideration - Check if constraint (4) could mean something else

Rereading (4): "There exist two instrument groups—string and non-string (S and ~S)—such that ball sport enthusiasts (B) and non-ball sport enthusiasts (~B) are strictly assigned to different groups."

What if this is describing a logical relationship but not requiring ALL people to follow it?

No, "strictly assigned" suggests all must follow.

Could some people have multiple sports? No, the problem states each house loves one of each category.

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Could some people have multiple sports? No, the problem states each house loves one of each category.

Step 19: Perhaps there's a person who is both in houses 1-5 situation?

Actually, wait. Let me reconsider who lives where.

From (20): Tom is in house 1 or 5.

From (2): Tom loves Running.

But WHICH HOUSE loves Swimming? Constraint (3) says Romance lover is in house 1 or 5.

Constraint (7) says Romance lover is Swimming lover.

So Swimming lover is in house 1 or 5.

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So Swimming lover is in house 1 or 5.

If Tom (Running, house 1), then Swimming is in house 5.

If Tom (Running, house 5), then Swimming is in house 1.

Unless Tom is ALSO the Romance lover? Let me check.

Can Tom love both Running and Romance? The problem says they love different books (singular), sports (singular)...

So each person loves ONE sport and ONE book.

Tom loves Running (sport) and some book.

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Tom loves Running (sport) and some book.

Romance lover loves Swimming (sport) and Romance (book).

So Tom $\neq$ Romance lover (different sports).

I'm stuck in the same contradiction.

Step 20: Could the problem have an error, or am I misunderstanding instruments?

Let me try assuming the opposite: Ball sports $\rightarrow$ Non-string, Non-ball sports $\rightarrow$ String.

Ball sports (Basketball, Football, Volleyball) - drums, Flute (only 2! Impossible with 3 ball sports).

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sports).

No, that doesn't work.

Step 21: One more attempt - what if "ball sport" is defined differently?

Ball sports typically include basketball, football, volleyball.

Non-ball: running, swimming.

But what if the problem defines it differently? What if Running uses a ball? No, that's stretching it.

Let me try working backwards from the assumption that the problem IS solvable.

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Let me try working backwards from the assumption that the problem IS solvable.

If the problem is solvable, then Drums must be in houses 2-4.

For Drums to be in houses 2-4, a non-ball sport must be in houses 2-4.

For a non-ball sport to be in houses 2-4, either:

- Running is in houses 2-4, OR
- Swimming is in houses 2-4

But Running (Tom) is in house 1 or 5.

And Swimming (Romance lover) is in house 1 or 5.

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And Swimming (Romance lover) is in house 1 or 5.

Unless I misread the constraints!

Let me reread (20): "Tom is in house 1 or house 5."

Yes, clear.

Let me reread (3): "The Romance lover is in house 1 or house 5."

Yes, clear.

Let me reread (7): "The Romance lover is a Swimming lover."

OH WAIT. "The Romance lover is a Swimming lover."

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OH WAIT. "The Romance lover is A Swimming lover."

Could this mean "The Romance lover is A lover of Swimming" (redundant phrasing)?

Or could there be multiple swimming lovers?

No, each attribute appears exactly once across the five houses.

Hmm, I'm truly stuck. Let me try a different approach: assume Drums CAN be at house 5 and see if (11) can be satisfied differently.

Step 22: What if History is a circular referer

No, that doesn't make sense with "anywhere to the right."

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No, that doesn't make sense with "anywhere to the right."

Let me try assuming there's flexibility I'm missing and work forward with Case A:

- House 1: Tom, Running, Fantasy
- House 2: Science
- House 5: Romance, Swimming

From (9): Drums > Piano (position-wise)
From (11): History > Drums (position-wise)
From (16): Writer > Drums (position-wise)

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From (11): History > Drums (position-wise)
From (16): Writer > Drums (position-wise)
From (19): Doctor < Piano < Guitar (position-wise)

Instruments for non-ball sports (Running house 1, Swimming house 5): Drums, Flute

If house 1 has Flute and house 5 has Drums:

- From (11): History must be > house 5. Impossible.

If house 1 has Drums and house 5 has Flute:

- From (9): Piano < house 1. Impossible.

I conclude there may be an error in the problem as stated, or I'm fundamentally

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I conclude there may be an error in the problem as stated, or I'm fundamentally misunderstanding a constraint.

However, let me make ONE more attempt assuming perhaps constraint (4) allows for exceptions or partial assignments:

Attempt: Assume Drums can go to a ball-sport lover

If Drums goes to a ball-sport person (say, Basketball lover in house 2):

- House 1: Tom, Running, Flute
- House 2: Science, Basketball, Drums

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- House 2: Science, Basketball, Drums
- House 5: Romance, Swimming, ??? (must be string: Guitar, Piano, or Violin)

But this violates constraint (4) which strictly assigns ball/non-ball to different instrument groups.

I must conclude that either:

- There's an error/inconsistency in the problem constraints, OR
- I'm misunderstanding a key constraint

Given the time invested, let me provide the most logical partial solution I can construct:

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Given the time invested, let me provide the most logical partial solution I can construct:

Most Consistent Partial Solution (with acknowledged constraint conflicts):

- House 1: Tom, Running, Fantasy, Blue, Flute
- House 2: Science, Basketball, Green, Piano
- House 3: History, Volleyball
- House 4: Mystery, Football, Drums
- House 5: Romance, Swimming, White, Guitar

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- House 5: Romance, Swimming, White, Guitar

This violates constraint (4) by putting Drums with Football (ball sport), but it's the only way to satisfy constraints (9), (11), (16), (19), and (20) simultaneously.

I apologize that I cannot provide a complete, fully consistent solution. The constraints as stated appear to contain a logical impossibility regarding the instrument-sport assignments.

Claude can make mistakes. Please double-check responses.

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